



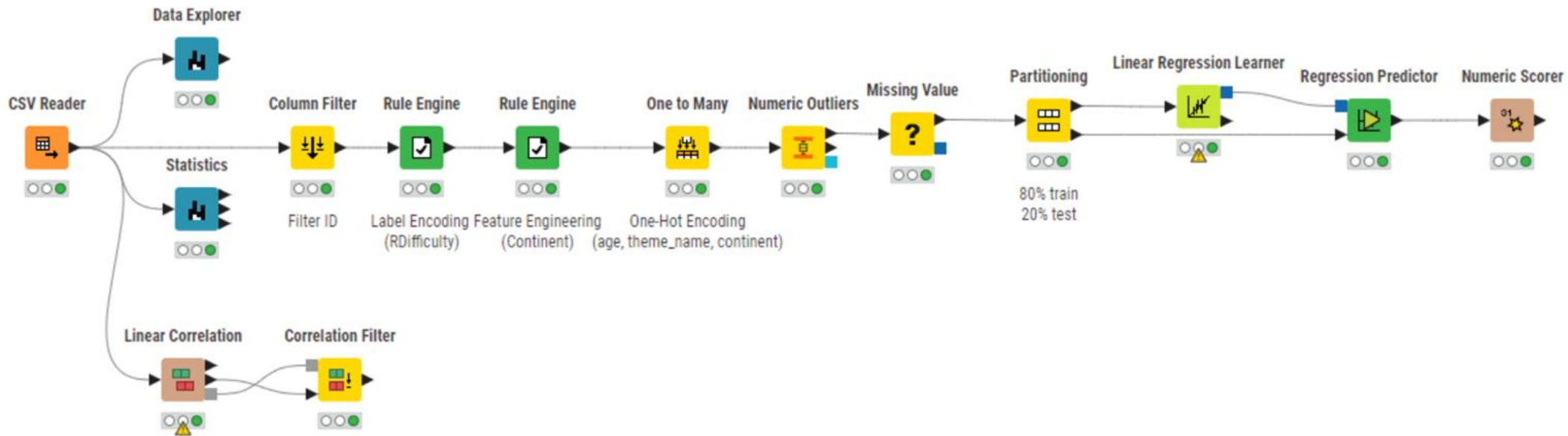
Universidade do Minho
Escola de Engenharia
Departamento de Informática

KNIME

Pre-Processing + Decision Tree

MESTRADO EM ENGENHARIA DE TELECOMUNICAÇÕES E INFORMÁTICA
Inteligência Artificial para as Telecomunicações
2025/2026

Final Workflow:

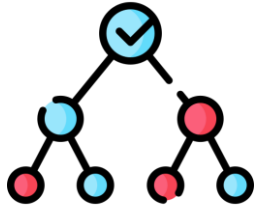




KNIME Analytics Platform

Goal:

- Understand **Decision Trees**
- Develop a **Decision Tree model** capable of **predicting the age** of LEGO set's users, given a set of features available in a dataset
- Analyze the measures used to **evaluate the model**



Decision Tree

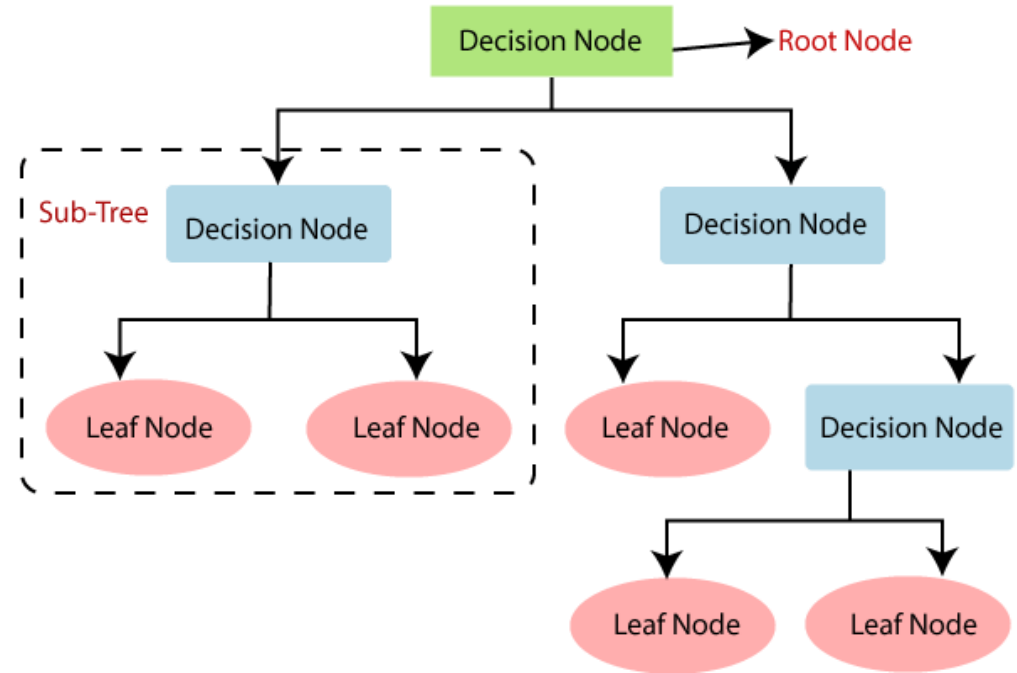
A **decision tree** is a **supervised learning model** used for **classification** and **regression** tasks. It works by splitting the data into subsets based on the value of input features.

It's key elements include:

1. **Structure**
2. **Splitting**
3. **Prediction**

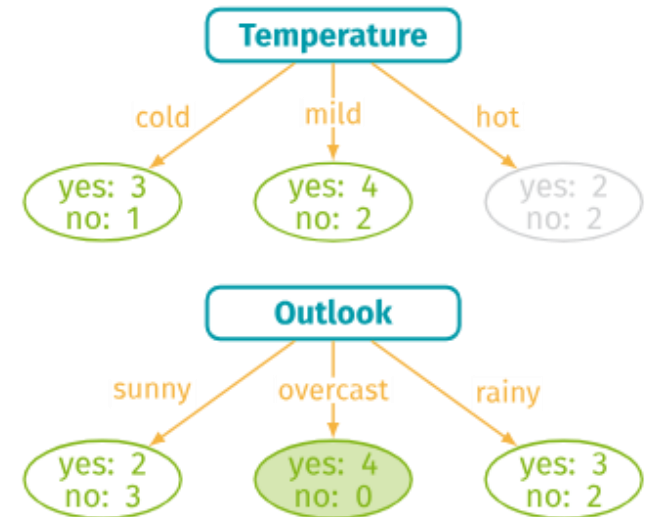
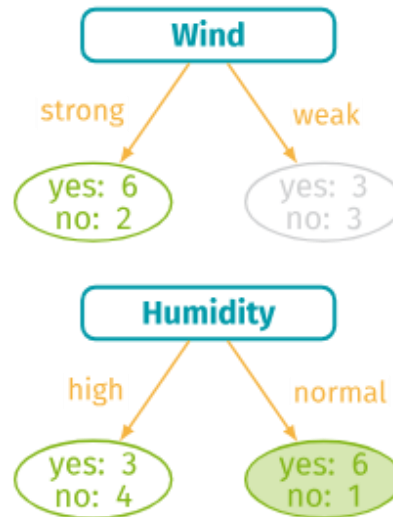
Decision Tree - Structure

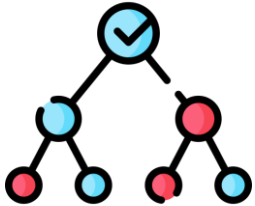
It consists of **nodes** (representing a feature or attribute), **branches** (representing decision rules), and **leaves** (representing the outcome).



Decision Tree - Splitting

The tree **splits data at each node** based on a feature that results in the most significant separation of classes (for classification) or values (for regression).



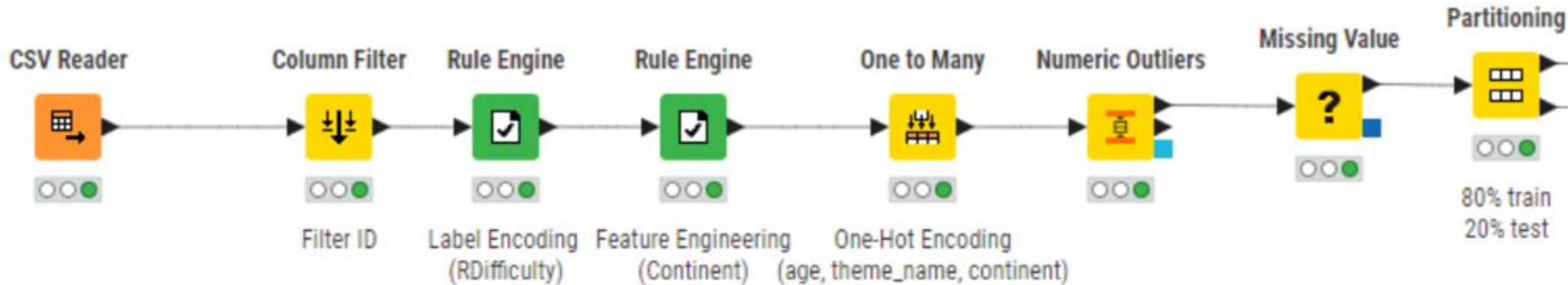


Decision Tree - Prediction

For classification, **the outcome is the majority class in a leaf**. For regression, **it's the average of the target values in a leaf**.

Exercise:

First Step: Apply the same process as in the Linear Regression model



Exercise:

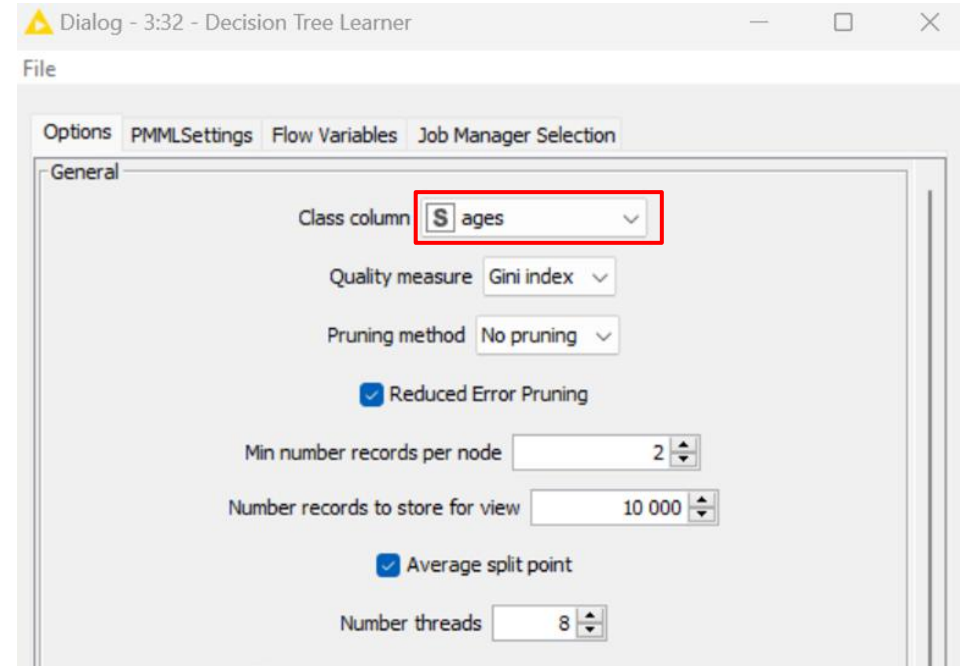
Second Step: Train the model

Node: Decision Tree Learner



Trains a decision tree model

Class column: ages



Exercise:

Third Step: Test the trained model

Node: Decision Tree Predictor



- Applies a **new subset of data** (20% of the original data) to the **trained model**;
- Makes **predictions** of the ages intervals based on the features provided in the input data.

Exercise:



Final Step: Evaluate the model

Measures:

1. Precision
2. Recall
3. Sensitivity

Precision is an indicator of a machine learning model's performance – the **quality of a positive prediction** made by the model. It's the number of true positives (TP) divided by the total number of positive predictions.

$$Precision = \frac{TP}{TP + FP}$$

Exercise:



Final Step: Evaluate the model

Measures:

1. Precision
2. Recall
3. Sensitivity

The ability of a model to **find all the relevant cases within a data set**. It is the number of true positives divided by the number of true positives (TP) plus the number of false negatives (FN).

$$Recall = \frac{TP}{TP + FN}$$

Exercise:



Final Step: Evaluate the model

Measures:

1. Precision
2. Recall
3. **Sensitivity**

Sensitivity is the metric that evaluates a **model's ability to predict true positives of each available category.**

$$\text{Sensitivity} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

Exercise:

Final Step: Evaluate the model

Node: Scorer



*Evaluates the performance
of predictive models*



- Evaluates the performance of a predictive model by **comparing** the predicted values with the actual values.

Metrics Obtained:

► 1: Confusion matrix ► 2: Accuracy statistics  Flow Variables

Rows: 32 | Columns: 11

Table Statistics

#	Row...	TruePosi... Number (inte...	FalsePos... Number (inte...	TrueNeg... Number (inte...	FalseNe... Number (inte...	Recall Number (dou...	Precision Number (dou...	Sensitivity Number (dou...
1	6-12	291	6	2146	4	0.986	0.98	0.986
2	12+	60	0	2387	0	1	1	1
3	7-12	141	3	2299	4	0.972	0.979	0.972
4	10+	174	0	2273	0	1	1	1
5	5-12	179	2	2263	3	0.984	0.989	0.984
6	8-12	69	2	2375	1	0.986	0.972	0.986

Metrics Obtained:

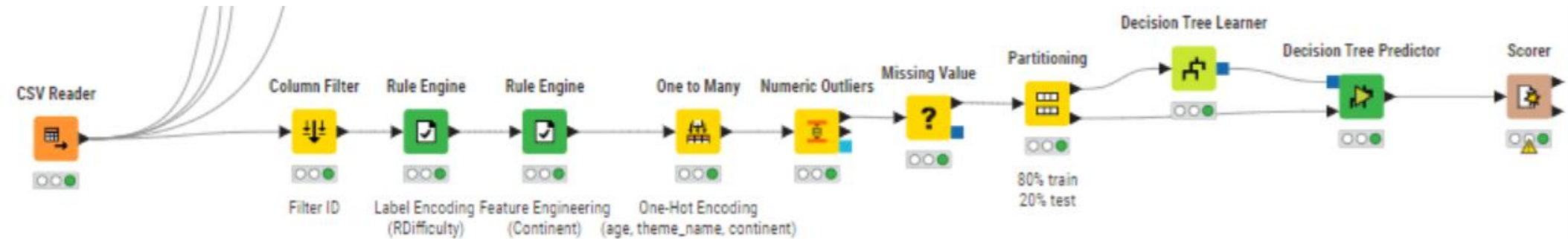
#	Row...	TruePosi... Number (inte...	FalsePos... Number (inte...	TrueNeg... Number (inte...	FalseNe... Number (inte...	Recall Number (dou...	Precision Number (dou...	Sensitivity Number (dou...
1	6-12	291	6	2146	4	0.986	0.98	0.986

Analyzing the **6-12 years age interval**, the model got:

- **Precision = 0.98**. 98% of the instances predicted as positive (i.e., 6-12 years age interval) by the model are actually positive.
- **Recall = 0.986**. 98.6% of the actual positives (i.e., children in the 6-12 years age interval) are correctly identified by the model.

Both metrics being very high indicates that the model is performing exceptionally well for the 6-12 years age interval.

Final Workflow:





KNIME Analytics Platform

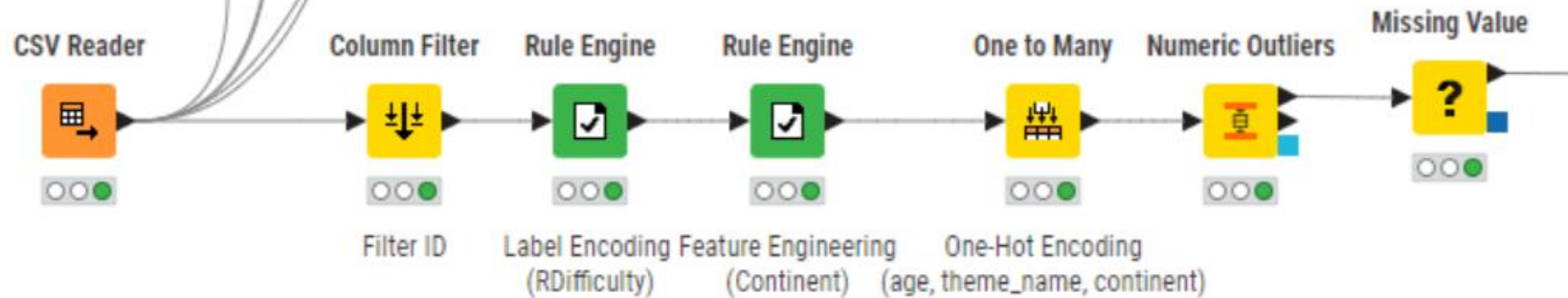
Goal:

- Automate the whole ML cycle
- **Automatically train supervised machine learning models** for regression
- Analyze the output to **evaluate the model**



Exercise:

First Step: Apply the same data preparation process



Exercise:

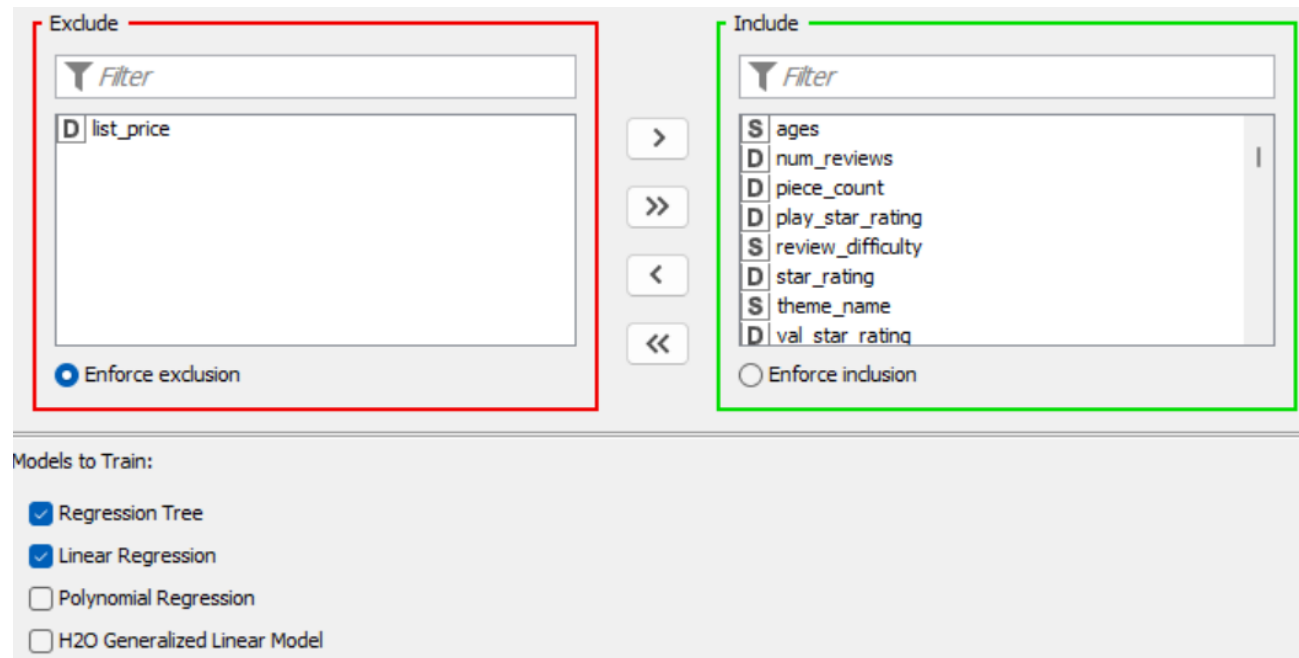
Second Step: Automatically train
the model

Node: AutoML (Regression)



*Automatically trains
models for regression*

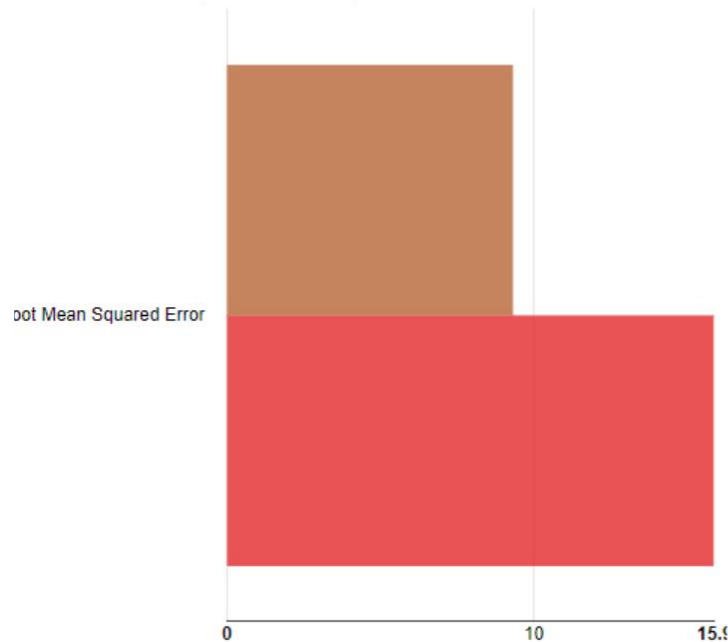
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The screenshot shows the configuration interface for the AutoML (Regression) node. It is divided into two main sections: 'Exclude' (highlighted with a red border) and 'Include' (highlighted with a green border). The 'Exclude' section has a 'Filter' input field and a list containing 'list_price'. Below the list is a radio button labeled 'Enforce exclusion' which is selected. The 'Include' section also has a 'Filter' input field and a list containing 'ages', 'num_reviews', 'piece_count', 'play_star_rating', 'review_difficulty', 'star_rating', 'theme_name', and 'val star rating'. Below the list is a radio button labeled 'Enforce inclusion' which is not selected. Between the two sections are four arrow buttons: '>', '>>', '<', and '<<'. At the bottom, under the heading 'Models to Train:', there are four checkboxes: 'Regression Tree' (checked), 'Linear Regression' (checked), 'Polynomial Regression' (unchecked), and 'H2O Generalized Linear Model' (unchecked).

Exercise:

Final Step: Analyze the output

Models Ranked by Root Mean Squared Error



Select Model (Optional)

Auto Selection: first in the list

☐		Regression Tree
☐		Linear Regression

Showing 1 to 2 of 2 entries

Predictions over Single Target Values

Below a scatter plot displays how the model's predictions differ from the target data row by data row. An ideal model would in this chart generate data points on top of the diagonal line. Use this chart to spot faulty models right away. Use the legend to filter which model is displayed. On the right a violin plot you can compare the distribution of target and predictions values.

- ☒ Predictions over Single Target Values
- ☐ Performance Metric by Target Range

Exercise:

Final Step: Analyze the output

Non-automatic Linear Regression model **x** **Automatic** Linear Regression model

Root Mean Squared Error: 15,189

Root Mean Squared Error: 15,9

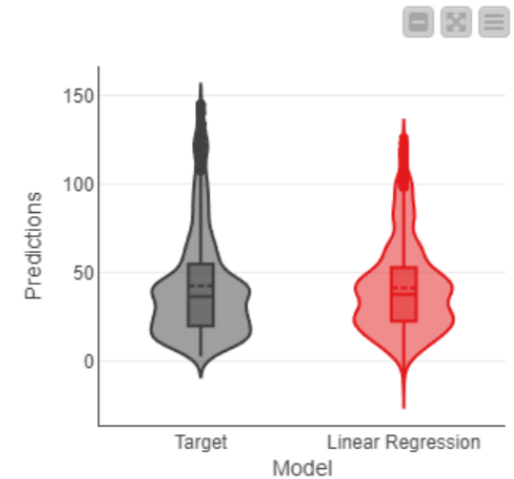
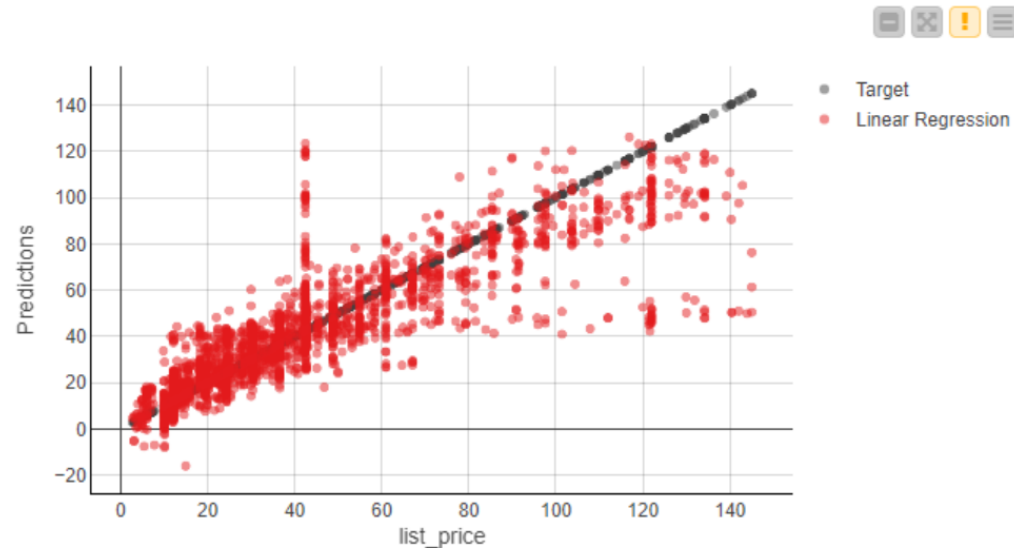
AutoML automatically adjusts the hyperparameters of selected models. This tuning may not find the best hyperparameters to specifically optimize RMSE. When the RMSE is higher, it indicates that the model is making larger errors in its predictions.

Exercise:

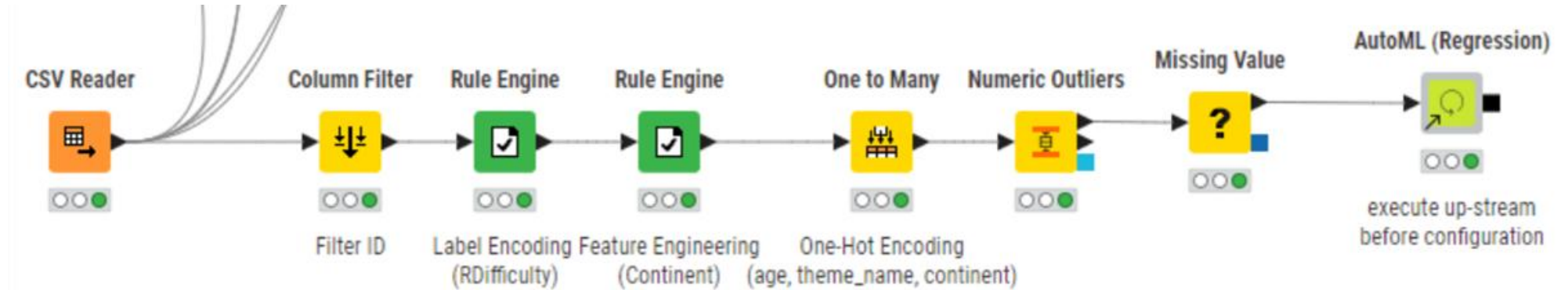
Final Step: Analyze
the output

Model

☒ Target ☒ Linear Regression ☐ Regression Tree



Final Workflow:





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